

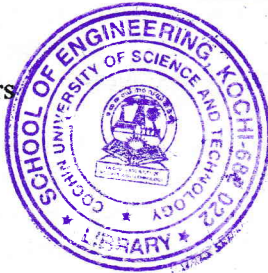
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B.Tech. Degree VI Semester Examination April 2019

ME 1604 HEAT AND MASS TRANSFER (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100



PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What is critical thickness of insulation? Derive an expression for critical thickness of insulation for a sphere.
- (b) Derive an expression for the temperature distribution for a lumped heat capacity system.
- (c) What is Reynold's analogy? Describe the relation between fluid friction and heat transfer.
- (d) Explain Fick's law of diffusion.
- (e) Define the following :
(i) Black body (ii) Gray body (iii) Irradiation (iv) Intensity of radiation
(v) Radiosity
- (f) What you mean by radiation shape factor? State and explain reciprocity theorem.
- (g) Explain about different types of heat exchangers.
- (h) Derive an expression for effectiveness of a parallel flow heat exchanger.

PART B

(4 × 15 = 60)

- II. A steel pipe 3 cm in diameter having its outer surface at 200°C is placed in air at 30°C with heat transfer coefficient of 8.5 W/m²K. It is proposed to add insulation ($k = 0.07 \text{ W/mK}$) on its outer surface to reduce heat loss by 40%. Estimate the thickness of insulation required, if pipe temperature and heat transfer coefficient remain unchanged. (15)

OR

- III. A cylindrical steel ingot ($k = 45 \text{ W/mK}$), 15 cm in diameter and 40 cm long passes through a treatment furnace, which is 6 m in length. The temperature of furnace gas is 1300°C. The initial ingot temperature is 100°C. The combined radiant and convective heat transfer coefficient is 100 W/m². Calculate the maximum speed with which ingot should pass through the furnace, if it must attain a temperature of 850°C. Take $\alpha = 0.46 \times 10^{-5} \text{ m}^2/\text{s}$. (15)

- IV. A vertical door of hot oven is 0.5 m high and is maintained at 200°C. It is exposed to atmospheric air at 20°C. Find (i) local heat transfer coefficient half way up the door (ii) average heat transfer coefficient for entire door (iii) thickness of the free convection boundary layer at the top of the door. (15)

OR

(P.T.O.)

- V. (a) Explain pool boiling and the different regimes of pool boiling with suitable sketches. (10)
 (b) Differentiate film wise and drop wise condensation. Which is more effective mechanism of heat transfer? Why? (5)
- VI. Two parallel infinite gray surfaces are maintained at temperature of 127°C and 227°C respectively. If the temperature of the hot surface is increased to 327°C , by what factor is the net radiation exchange per unit area increased? Assume the emissivities of colder and hotter surfaces to be 0.9 and 0.7 respectively. (15)
- OR**
- VII. Two large parallel planes with emissivity 0.6 are at 900 K and 300 K. A radiation shield with one side polished and having emissivity of 0.05, while the emissivity of other side is 0.4 is proposed to be used. Which side of the shield to face the hotter plane, if the temperature of the shield to be kept minimum? Justify your answer. (15)
- VIII. Hot oil with a capacity rate of 2500 W/K flows through a double pipe heat exchanger. It enters at 360°C and leaves at 300°C . Cold fluid enters at a temperature of 30°C and leaves at 200°C . If the overall heat transfer coefficient is 800 W/m^2 , determine heat exchanger area required for (i) parallel flow (ii) counter flow. (15)
- OR**
- IX. Calculate for the following cases, the surface area required for a heat exchanger which is required to cool 3200 kg/hr of benzene ($C_p = 1.74 \text{ kJ/kg}^{\circ}\text{C}$) from 72°C to 42°C . The cooling water ($C_p = 4.18 \text{ kJ/kg}^{\circ}\text{C}$) at 15°C has a flow rate of 2200 kg/hr. (15)
 (i) Single pass counter-flow.
 (ii) 1-4 exchanger (one-shell pass and four-tube passes).
